

Title

Cardiometabolic and inflammatory biomarker responses to structured exercise in older women: secondary analysis by metabolic syndrome status

Abstract

Background: Understanding how exercise modulates systemic cardiometabolic and inflammatory markers in older adults with varying metabolic status informs personalized intervention design. **Objective:** Secondary analysis of cardio-metabolic and inflammatory biomarker changes following 10-week supervised exercise. **Methods:** Body composition via densitometry, lipid fractionation, glucose regulation markers (fasting glucose and HOMA-IR), and inflammatory biomarkers (C-reactive protein, interleukin-6) were measured in 60 participants stratified by metabolic syndrome status. **Results:** Exercise participants demonstrated significant improvements across metabolic syndrome status: body fat decreased mean 3.1%, low-density lipoprotein cholesterol fell by 8%, triglycerides declined 11%, and HOMA-IR improved 12%. High-density lipoprotein cholesterol increased by 6% ($P = 0.003$). C-reactive protein and interleukin-6 concentrations decreased by 24% and 18%, respectively, in the intervention arms ($P < 0.05$). Improvements were slightly greater in metabolic syndrome participants on several measures, suggesting particular benefit in this phenotype. No significant changes occurred in the usual-activity control arm. **Conclusion:** Supervised exercise produces substantial improvements in cardiometabolic risk factors and inflammatory markers in older women, with notably favorable responses in those with established metabolic syndrome.